

Proton-Precession Magnetometer Physics calculations project details and Sensor-Head Design Notes

Physics & Math Behind the Coil Geometry and Sizing for a Target: 1 nT Resolution

1. What the sensor head must do

A proton-precession magnetometer measures the magnitude of the ambient magnetic field B_0 by measuring the proton Larmor frequency:

$$f_0 = \left(\frac{\gamma_p}{2\pi} \right) B_0, \quad \frac{\gamma_p}{2\pi} \approx 42.577 \times 10^6 \text{ Hz/T.} \quad (1)$$

To Measure:

1. **Polarize:** Apply a strong temporary field B_p to align nuclear spins.
2. **Switch off:** Turn B_p off quickly and wait a short *blanking* delay.
3. **Listen:** Record the decaying precession signal (FID) and estimate f_0 , then compute B_0 .

2. What 1 nT means

A 1 nT field change corresponds to

$$\Delta f_{1 \text{ nT}} = \left(\frac{\gamma_p}{2\pi} \right) 1 \times 10^{-9} \text{ T} \approx 0.0426 \text{ Hz.} \quad (2)$$

So achieving 1 nT resolution requires reliably estimating frequency to roughly $4 \times 10^{-2} \text{ Hz}$

3. Polarization physics

In thermal equilibrium at temperature T , nuclear polarization is small and scales linearly with B_p :

$$p \approx \frac{\mu_p B_p}{kT}, \quad (3)$$

so the initial magnetization increases roughly linearly with B_p and with sample volume V_s :

$$M_{\perp}(0) \propto p V_s \propto B_p V_s. \quad (4)$$

The magnetization builds during the polarization interval T_{pol} with a T_1 time constant:

$$M(T_{\text{pol}}) = M_{\text{eq}} \left(1 - e^{-T_{\text{pol}}/T_1} \right). \quad (5)$$

What this means for our groups Design: Increasing B_p and/or using a larger sample increases SNR; choosing a fluid with shorter T_1 allows shorter cycles for the same magnetization.

4. Precession signal model

After switch-off, the transverse magnetization precesses about B_0 and decays with T_2^* :

$$M_{\perp}(t) = M_{\perp}(0) e^{-t/T_2^*} \cos(\omega_0 t + \phi), \quad \omega_0 = 2\pi f_0. \quad (6)$$

A pickup coil converts changing flux into voltage:

$$v(t) = -N_s \frac{d\Phi(t)}{dt} \Rightarrow \text{(scaling)} \quad v_{\text{sig}} \propto N_s \omega_0 M_{\perp}(0) \text{ (coupling)}. \quad (7)$$

Signal improves with B_p , sample volume, pickup turns N_s , and good geometric coupling.

5. Polarizing coil sizing

5.1 Field at center of a finite solenoid

For a solenoid of length ℓ , radius a , and N turns carrying current I , the on-axis field at the center can be approximated by:

$$B_p(0) \approx \mu_0 \frac{N}{\ell} I \cdot k, \quad k = \frac{\ell}{2\sqrt{a^2 + (\ell/2)^2}} = \frac{1}{\sqrt{1 + (2a/\ell)^2}}. \quad (8)$$

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5.2 Ampere-turns requirement

Rearranging our previous equation gives the required ampere-turns for a target B_p :

$$NI \approx \frac{B_p \ell}{\mu_0 k}. \quad (9)$$

Tradeoffs:

- Increasing N increases B_p for fixed I , but increases resistance R and inductance L .
- Increasing I increases B_p linearly, but increases heating $P = I^2 R$.

5.3 Heating and duty cycle

$$P_{\text{coil}} = I^2 R, \quad E_{\text{pulse}} = I^2 R T_{\text{pol}}. \quad (10)$$

Design implication: We will need to Use thicker wire for a lower R if we plan to use high current without excessive heating.

6. Switch-off physics

The polarizer is an inductor; rapid switch-off uses voltage:

$$V \approx L \frac{\Delta I}{\Delta t}. \quad (11)$$

and stores energy:

$$E_L = \frac{1}{2} L I^2, \quad (12)$$

which must be safely dissipated or recovered every pulse. More turns can raise L substantially however it may force large switch-off voltages.

7. Pickup coil design (sensitivity vs noise)

7.1 Johnson noise

A dominant fundamental noise source is thermal (Johnson) noise of the pickup coil resistance:

$$e_n \approx \sqrt{4kTR\Delta f} \quad [\text{V}_{\text{rms}}], \quad (13)$$

where Δf is measurement bandwidth.

This essentially means Increasing turns N_s raises signal but also increases resistance R raising. Most Practical DIY designs choose a wire gauge that balances these.

7.2 Gradiometer/coil cancellation

A single pickup coil is an antenna for environmental magnetic noise. A practical high-sensitivity approach uses two identical coils:

- **Sample coil:** contains the bottle, sees proton signal + environmental noise.
- **Dummy coil:** identical geometry/winding, no bottle, sees environmental noise only.

Wiring them in series opposition cancels common mode environmental pickup while preserving the proton signal. Matching turns, lead routing, potting, and cable is critical.

8. Orientation

The usable transverse magnetization after switch-off depends on the angle α between the polarization direction and the ambient field direction:

$$M_{\perp}(0) \propto M(T_{\text{pol}}) \sin \alpha. \quad (14)$$

By using a tilt mount we will be able to essentially choose α to avoid alignment that would reduce signal amplitude.

9. Our chosen baseline geometry (buildable + nT-capable)

The following geometry is a practical starting point that supports nT-class performance when paired with good switching/blanking and a stable timebase.

9.1 Polarizing coil

Target: $B_p \approx 7.5 \text{ mT}$ (the needed baseline), with optional headroom for higher B_p if switching allows.

Parameter	Value
Coil inner diameter (ID)	11.4 cm
Coil length ℓ	10.0 cm
Turns N	228 (6 layers $\times$ 38 turns/layer)
Radial winding thickness	1.43 cm
Wire	14 AWG solid copper (insulated)
Electrical (ballpark)	$L \approx 4.3 \text{ mH}$, $R \approx 1.1 \Omega$
Former material	Acrylic tube + acrylic end plates

9.2 Pickup coils (sample + dummy)

Parameter	Value
Coil inner diameter (ID)	3.8 cm
Coil length	10.0 cm
Turns N_s (each coil)	552 (4 layers $\times$ 138 turns/layer)
Wire	22 AWG magnet wire
Electrical (ballpark)	$L \approx 3.5$ mH, $R \approx 5.9 \Omega$ (incl. cable)
Configuration	Two identical coils; bottle in one coil only

10. How this supports a 1 nT goal

1. **Frequency target:** 1 nT $\leftrightarrow$ 0.0426 Hz.
2. **SNR:** Achieved by sufficiently large B_p , good fill factor , and cancellation (dummy coil).
3. **Usable time:** Keep T_2^* large by magnetic cleanliness and stable mechanics; minimize blanking so you use early high-SNR FID.
4. **Switch-off:** Ensure the polarizer clamp/switch can handle $V \sim L\Delta I/\Delta t$ and energy E_L .
5. **Timebase/estimator:** Use a stable clock and a frequency estimator better than naive 1-second counting (may be constrained by budget).